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Studierichting: Informatica
Oefeningenreeks 5A nr. 11.1

$$\int_0^{\frac{\pi}{2}} x^2 \sin x dx$$

$$\begin{cases} u = x^2 \\ dv = \sin x dx \end{cases} \Rightarrow \begin{cases} du = 2x dx \\ v = -\cos x \end{cases}$$

$$= \left[-x^2 \cos x\right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} 2x \cos x dx$$

$$= \left[-x^2 \cos x\right]_0^{\frac{\pi}{2}} + 2 \int_0^{\frac{\pi}{2}} x \cos x dx$$

$$\begin{cases} u = x \\ dv = \cos x dx \end{cases} \Rightarrow \begin{cases} du = dx \\ v = \sin x \end{cases}$$

$$= \left[-x^2 \cos x\right]_0^{\frac{\pi}{2}} + 2 \left([x \sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x dx \right)$$

$$= \left[-x^2 \cos x\right]_0^{\frac{\pi}{2}} + 2 \left([x \sin x]_0^{\frac{\pi}{2}} - [-\cos x]_0^{\frac{\pi}{2}} \right)$$

$$= \left(-\frac{\pi^2}{4} \cos \frac{\pi}{2} - 0 \right) + 2 \left(\frac{\pi}{2} \sin \frac{\pi}{2} - \left(-\cos \frac{\pi}{2} - (-\cos 0)\right) \right)$$

$$= 0 + 2 \left(\frac{\pi}{2} - 1 \right)$$

$$= \pi - 2$$

References